Executive Summary

AI-Driven Data Research in Quantitative Finance

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Project Overview

This executive summary presents the IEDA4920 Final Year Project, *AI-Driven Data Research in Quantitative Finance*. The project develops an end-to-end framework that converts SEC Item 1A *Risk Factors* disclosures into hierarchical textual-risk exposures, semantic peer networks, topology-only ST-GAT forecasts, econometric diagnostics, and academic portfolio-backtesting outputs.

Annual 10-K risk paragraphs are extracted from EDGAR filings, mapped into a dynamic macro-meso taxonomy, aggregated into firm-year exposure vectors, and converted into annual semantic peer graphs. These graphs are used to test whether disclosure-implied peer topology improves one-year-ahead return and volatility forecasts beyond each firm's own financial and market features.

Core Research Contribution

The central modeling contribution is a strict graph-topology ablation. The full ST-GAT model uses

$$A_t = I + A_{risk,t},$$

while the identity baseline uses

$$A_t = I.$$

This design isolates the incremental value of textual-risk peer topology. If the full graph improves over the identity baseline, the gain is attributable to semantic peer links rather than to a different feature set or a weaker model family.

Empirical Design

The forecasting panel covers risk years 2006–2024. The chronological split is:

Stage	Years	Role
Training	2006–2016	Model fitting, feature scaling, and imputation statistics
Validation	2017–2020	Hyperparameter selection and early-stopping signal
OOS review panel	2021–2024	Final forecast and portfolio diagnostics

Each firm-year node uses 12 backward-looking financial and market features, standardized using training-year statistics only. Features and risk topology are aligned to the July-to-June feature window, while forward return and volatility targets are measured over the following July-to-June window.

Main Verified Results

The finalized report is conservative about the default Meso graph evidence. On the exported 2021–2024 OOS review panel, the default Meso ST-GAT improves return MAE and return rank, but not averaged OOS RMSE. A full-window `theta040` taxonomy sensitivity run provides stronger forecast-error evidence:

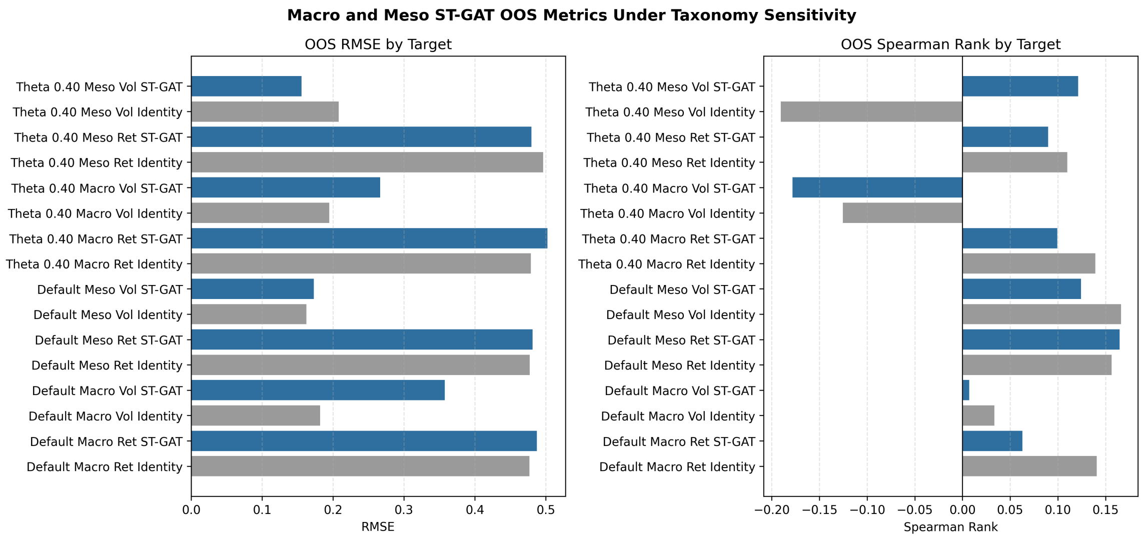
Metric	Identity baseline	ST-GAT
Averaged RMSE (`theta040`)	0.3523	0.3176
Averaged MAE (`theta040`)	0.2546	0.2012
Averaged Spearman rank (`theta040`)	-0.0403	0.1055
Volatility RMSE (`theta040`)	0.2082	0.1555
Default return Spearman rank	0.1564	0.1650

The clearest improvement is in volatility forecasting under `theta040`. Paired Macro-level runs are much denser but underperform identity baselines, so the report interprets the result as selective Meso information transfer rather than a generic benefit from broad category coverage. The default taxonomy remains useful as a conservative baseline because it shows that graph gains are not automatic and must be sensitivity-tested.

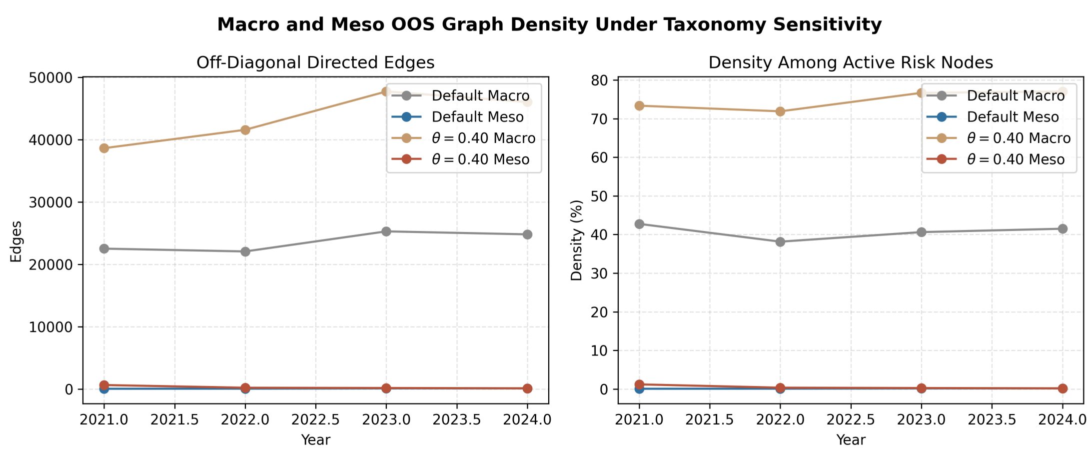
Graph Interpretation

Graph-density diagnostics support the interpretation that taxonomy sensitivity matters. During the OOS period, the default Meso graph retains 32–96 off-diagonal directed edges per year. The `theta040` Meso graph retains 100–630 edges and is materially denser in 2021, while Macro graphs retain tens of thousands of links and become overconnected.

The report therefore frames the graph result as conditional evidence: taxonomy settings alter graph density, forecast behavior, and portfolio ranking.



OOS forecasting comparison. The report compares Macro and Meso default and `theta040` outputs against matched identity baselines.



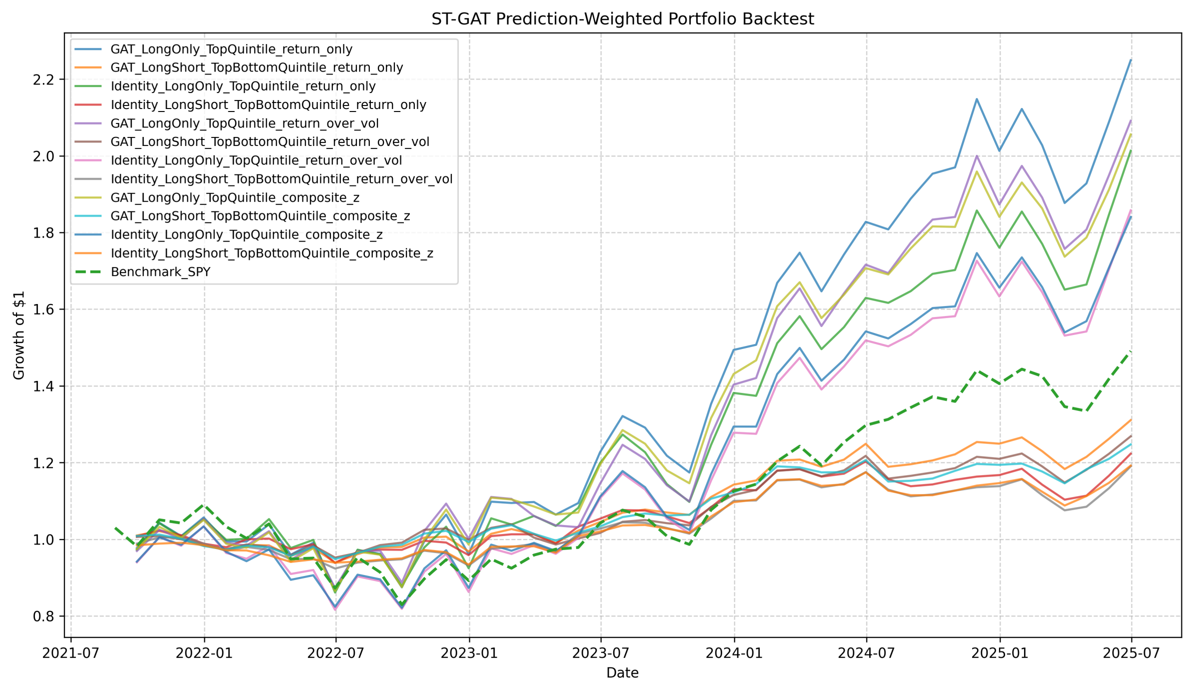
Graph-density diagnostic. Macro and Meso taxonomies produce very different OOS graph densities.

Portfolio Validation

The portfolio extension is presented as academic validation, not as live-trading evidence. Annual Meso ST-GAT forecasts are held fixed over their July-to-June forecast window, while portfolios are rebalanced monthly with 10 bps transaction costs, 5 bps slippage, and SPY as benchmark.

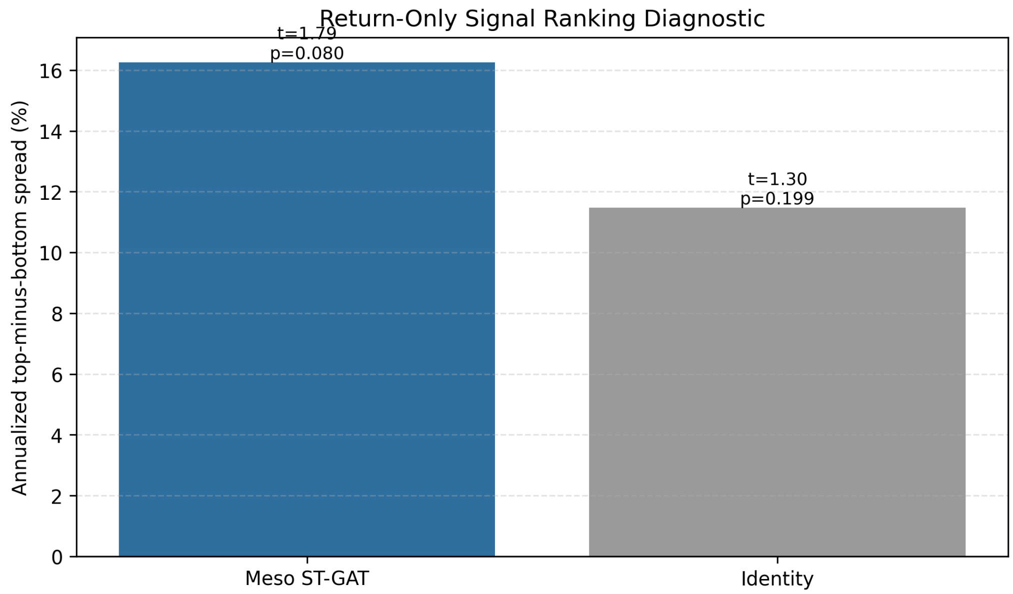
The strongest return-only long-short strategy reports:

Measure	Value
Annualized return	7.31%
Annualized volatility	8.34%
Sharpe ratio	0.88
Sortino ratio	1.52
Maximum drawdown	-8.06%
Beta versus SPY	0.30
Top-minus-bottom spread	16.26% annualized
Nominal spread p-value	0.080



Backtest comparison. Monthly rebalanced ST-GAT and identity-baseline portfolio variants are compared against SPY using the exported equity curve.

The identity-baseline return-only signal has a smaller and insignificant 6.62% annualized top-minus-bottom spread. This supports the interpretation that semantic peer topology improves cross-sectional ranking quality, while the report avoids presenting the result as investment advice or live-trading deployability.



Portfolio ranking diagnostic. The Meso ST-GAT return-only signal produces the strongest top-minus-bottom spread in the current academic backtest outputs.

Claim Safety and Limitations

The final thesis is deliberately conservative about unsupported evidence. Old taxonomy-coverage percentages, unsupported single-firm migration claims, DAV/EGARCH-X BIC comparisons, Fama-MacBeth tables, and SAR/network regression claims are not used as headline findings unless their source tables can be independently audited. The reported taxonomy coverage and LVS dynamic case-study figures are backed by auditable source tables and are interpreted as taxonomy diagnostics, not standalone proof of forecast value.

The available DAV peak diagnostics are retained only as event-local context: 93 figures were parsed, with 20 positive improvements, 73 negative improvements, and mean improvement of -2.40%. These diagnostics do not establish broad DAV dominance.

Validation Status

The empirical evidence passed the validation audit with 35 passes, 7 warnings, and 0 failures as of 2026-05-03. The warnings are research caveats rather than model failures: optional multi-seed, DAV, and taxonomy-interim evidence is not used as headline evidence; four financial feature values require train-period median imputation; and risk-year 2024 has a forward target window ending on 2026-06-30.